

LAB 2:

This lab introduced basic ROS 2 concepts using the Turtlesim simulator. First, the ROS 2 environment was set up and the turtlesim node was launched. A separate teleoperation node was used to control the turtle with the keyboard. The turtle responded to user inputs in real time, confirming proper communication between nodes.

Next, ROS 2 topics were explored. Active topics were listed, and the turtle's position was monitored using the pose topic. Continuous updates showed how data is published and subscribed between nodes. Velocity commands were then sent manually using the command velocity topic. The turtle moved according to the specified linear and angular values.

Services were tested by resetting the simulation. The reset service returned the turtle to its initial position, showing how services perform specific tasks on request.

The rqt tool was used to visualize the ROS system. Through rqt, services like reset and spawn were called using a graphical interface. A second turtle was successfully spawned and controlled independently using its own command velocity topic. This demonstrated multi-node communication and independent control.

Observations:

The turtle moved smoothly when given velocity commands.

Real time position updates were accurate and continuous.

Services executed instantly and affected the simulation as expected.

The second turtle operated independently without affecting the first.

This lab helped build a clear understanding of nodes, topics, and services in ROS 2, and showed how different components interact in a robotic system.